

Unit 5: Clusters

Cluster Sampling

- A. Clusters and survey costs
- B. Variance of the mean
- C. Design effect

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A. Cluster and survey costs

- Populations often distributed geographically
 - Cannot afford to visit n units drawn randomly from the entire area
 - Cannot afford to create an element frame
- Cluster selections are used to reduce costs
 - Select clusters and list elements only for selected clusters
- Clusters naturally occurring units:
 - Seldom equal size

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B. Variance of the mean

Target Population
Elements:

101 Main St.

104 Main St.

107 Main St.

112 Main St.

115 Main St.

122 Main St.

129 Main St.

132 Main St.

201 Main St.

206 Main St.

.

.

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$$\bar{Y} = \sum_{i=1}^N Y_i / N$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N - 1)$$

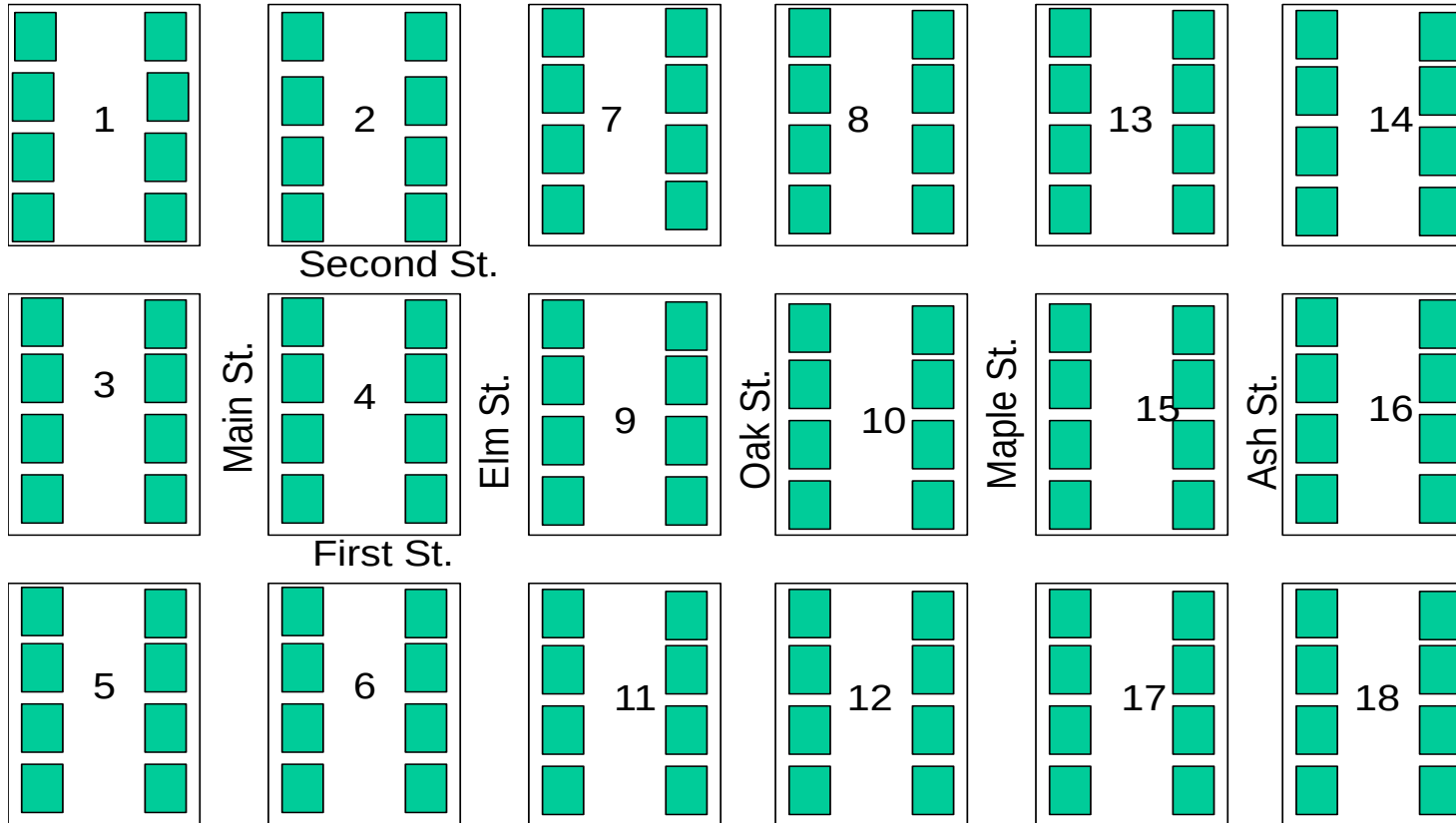
Notation

- A = the number of clusters in the population
- a = the number of clusters in the sample
- B = the number of elements in each cluster
- $\frac{a}{A}$ is the rate at which clusters are selected
- If we select all elements within a cluster, the rate is $\frac{B}{B} = 1$
- $\frac{a}{A} \times \frac{B}{B} = \frac{n}{N}$

Clustered population

$$\bar{Y}_\alpha = \sum_{\beta=1}^B Y_{\alpha\beta} / B$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



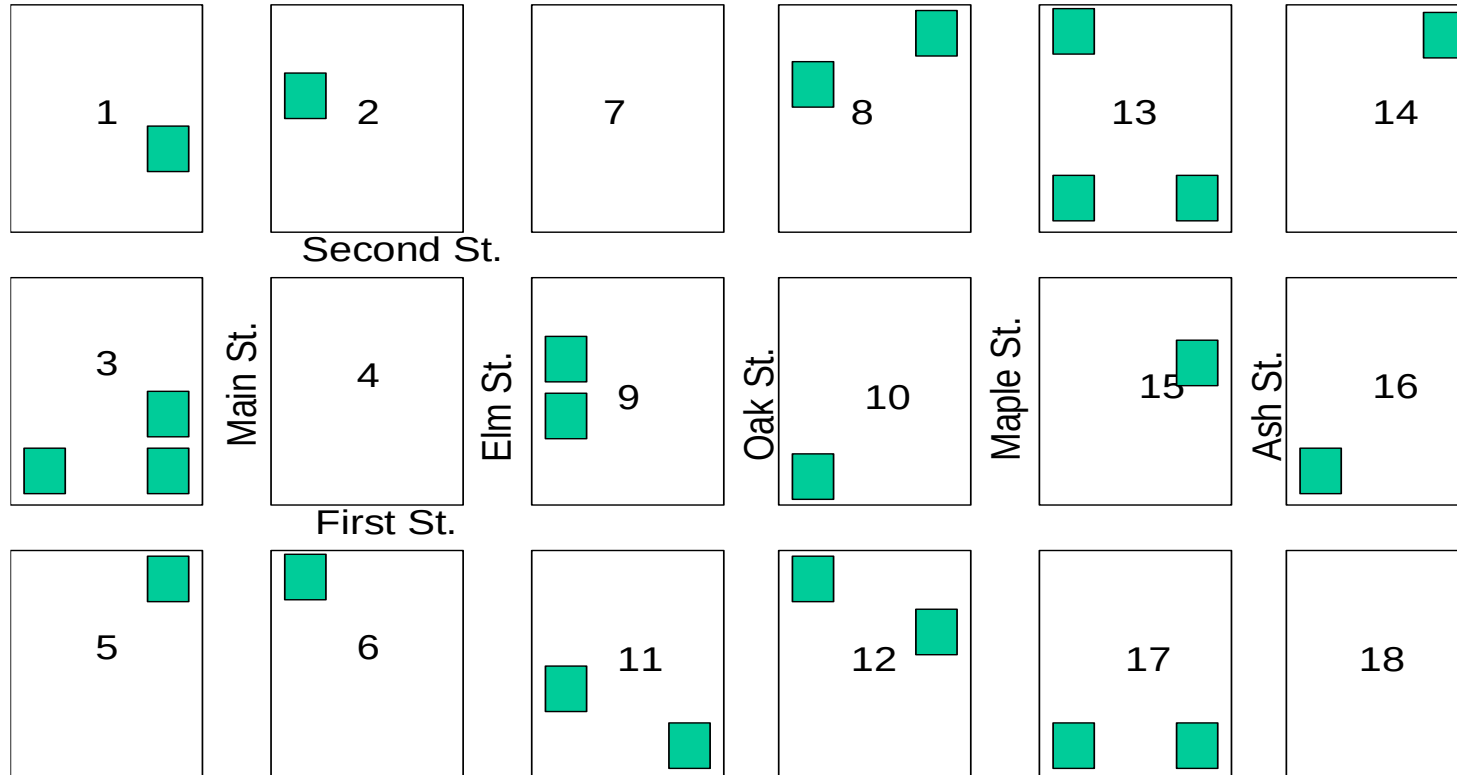
$$S_\alpha^2 = \sum_{\beta=1}^B (Y_{\alpha\beta} - \bar{Y}_\alpha)^2 / (B-1)$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Element sample

$$\bar{y} = \sum_{i=1}^n y_i / n$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$s^2 = \sum_{i=1}^n (y_i - \bar{y})^2 / (n-1)$$

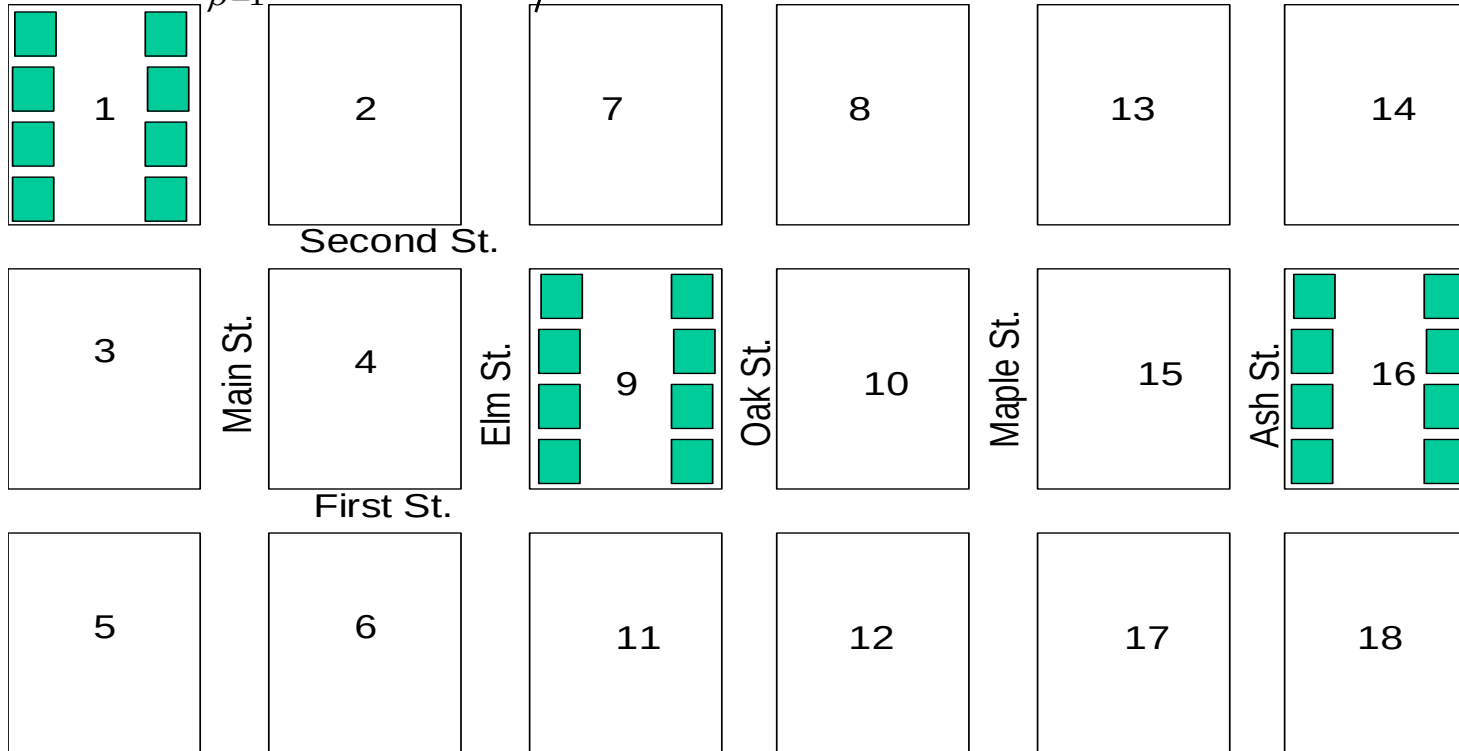
$$Var(\bar{y}) = \frac{(1-f)}{n} S^2$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Cluster sample

$$\bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^B y_{\alpha\beta} / aB \quad s_{\alpha}^2 = \sum_{\beta=1}^B (y_{\alpha\beta} - \bar{y}_{\alpha})^2 / (B-1)$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$s_a^2 = \sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2 / (a-1)$$

$$Var(\bar{y}) = \frac{(1-f)}{a} S_a^2$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Computing formulas

- Here we estimate

$$\text{var}(\bar{y}) = \frac{1-f}{a} s_a^2 = \frac{1-f}{a} \frac{\sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2}{a-1}$$

- But there is an algebraic expansion...

$$\text{var}(\bar{y}) = \frac{1-f}{a} \frac{\left[\sum_{\alpha=1}^a \bar{y}_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a \bar{y}_{\alpha} \right)^2}{a} \right]}{a-1}$$

Computing formulas

- And that algebraic expansion is converted into one involving cluster totals:

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{a} \left[\frac{\sum_{\alpha=1}^a \bar{y}_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a \bar{y}_{\alpha}\right)^2}{a}}{a-1} \right] = \frac{1-f}{a(a-1)} \left[\sum_{\alpha=1}^a \left(\frac{y_{\alpha}}{B}\right)^2 - \frac{\left(\sum_{\alpha=1}^a \frac{y_{\alpha}}{B}\right)^2}{a} \right] \\ &= \frac{1-f}{aB^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha}\right)^2}{a} \right]\end{aligned}$$

Computing formulas

- So, for taking complete clusters (simple cluster sampling),

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{a} s_a^2 \\ &= \frac{1-f}{a} \frac{\sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2}{a-1} \\ &= \frac{1-f}{aB^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right]\end{aligned}$$

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C. Design effect

- There is no particular reason $\text{var}(\bar{y}) \neq \text{var}_{SRS}(\bar{y})$
- For cluster sampling:

$$deff = \frac{\text{Var}(\bar{y})}{\text{Var}_{srs}(\bar{y})} = \frac{\frac{1-f}{a} S_a^2}{\frac{1-f}{n} S^2} = \frac{S_a^2 / a}{S^2 / n} > 1.0$$

Design effect: Alternate form

$$\begin{aligned} \text{Var}(\bar{y}) &= \text{Var}\left(\frac{1}{n} \sum_{\alpha} \sum_{\beta} y_{\alpha\beta}\right) \\ &= \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} \text{Var}(y_{\alpha\beta}) + \sum_{\alpha, \alpha'} \sum_{\beta, \beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha'\beta'}) \right] \\ &= \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} \text{Var}(y_{\alpha\beta}) + \sum_{\alpha \neq \alpha'} \sum_{\beta} \sum_{\beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha'\beta'}) + \sum_{\alpha} \sum_{\beta \neq \beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha\beta'}) \right] = \\ &\quad \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} S^2 + \sum_{\alpha \neq \alpha'} \sum_{\beta} \sum_{\beta'} 0 + \sum_{\alpha} \sum_{\beta \neq \beta'} \rho S^2 \right] \\ &= \frac{1}{n^2} \left[nS^2 + a \cdot B(B-1)\rho S^2 \right] = \frac{S^2}{n} \left[1 + (B-1)\rho \right] \end{aligned}$$

Design effect: Intra-cluster homogeneity

- Hence

$$\begin{aligned} Deff &= \frac{Var(\bar{y})}{Var_{srs}(\bar{y})} \\ &= \frac{\frac{S^2}{n} [1 + (B-1)\rho]}{\frac{S^2}{n}} \\ &= [1 + (B-1)\rho] \end{aligned}$$

Features of ρ

- $-\frac{1}{(B-1)} < \rho < 1$ (although $\rho > 0$ generally)
- If $\rho = 0$, $Deff = 1.0$
 - The cluster sample is the equivalent of SRS of size $n = a \times B$
- If $\rho = 1$, $Deff = B$ and $Var(\bar{y}) = B \times Var_{srs}(\bar{y})$
 - The cluster sample is equivalent to an SRS of a elements

Source and nature of ρ

- For human populations in naturally occurring clusters, factors include
 - Environment (exposure to infectious disease)
 - Self-selection (poor households in same block)
 - Interaction (shared attitudes among neighbors)
- The size depends on
 - Characteristic Y (disease status, age)
 - Nature of clusters (naturally formed)
 - Size of cluster (blocks, census tracts, counties)

Estimation of ρ

- Proper estimation, especially for multistage stratified samples, is cumbersome
- It is useful for design to estimate ρ in a straightforward fashion
- Synthetic estimate:

$$roh = \frac{deff - 1}{B - 1}$$

Sample size considerations

- Implications of $\rho > 0$ for sample size:
 - Compute an SRS sample size (n_{srs})
 - For B and ρ (“guesstimate?”), compute $n = \left[1 + (B-1)\rho\right] n_{srs}$
- Design effect and confidence intervals:
 - With $(a-1)$ d.f.: $\bar{y} \pm \left(t_{(1-\alpha/2; a-1)}\right) se(\bar{y})$

or

$$\bar{y} \pm \left(t_{(1-\alpha/2; a-1)}\right) \sqrt{\left[1 + (B-1)\rho\right] \frac{s^2}{n}}$$

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- B. Sample size and *roh*
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A. Subsampling clusters

- Select $b < B$ elements from a clusters *epsem* :

$$f = \frac{a}{A} \times \frac{b}{B} = \frac{a \times b}{A \times B} = \frac{n}{N}$$

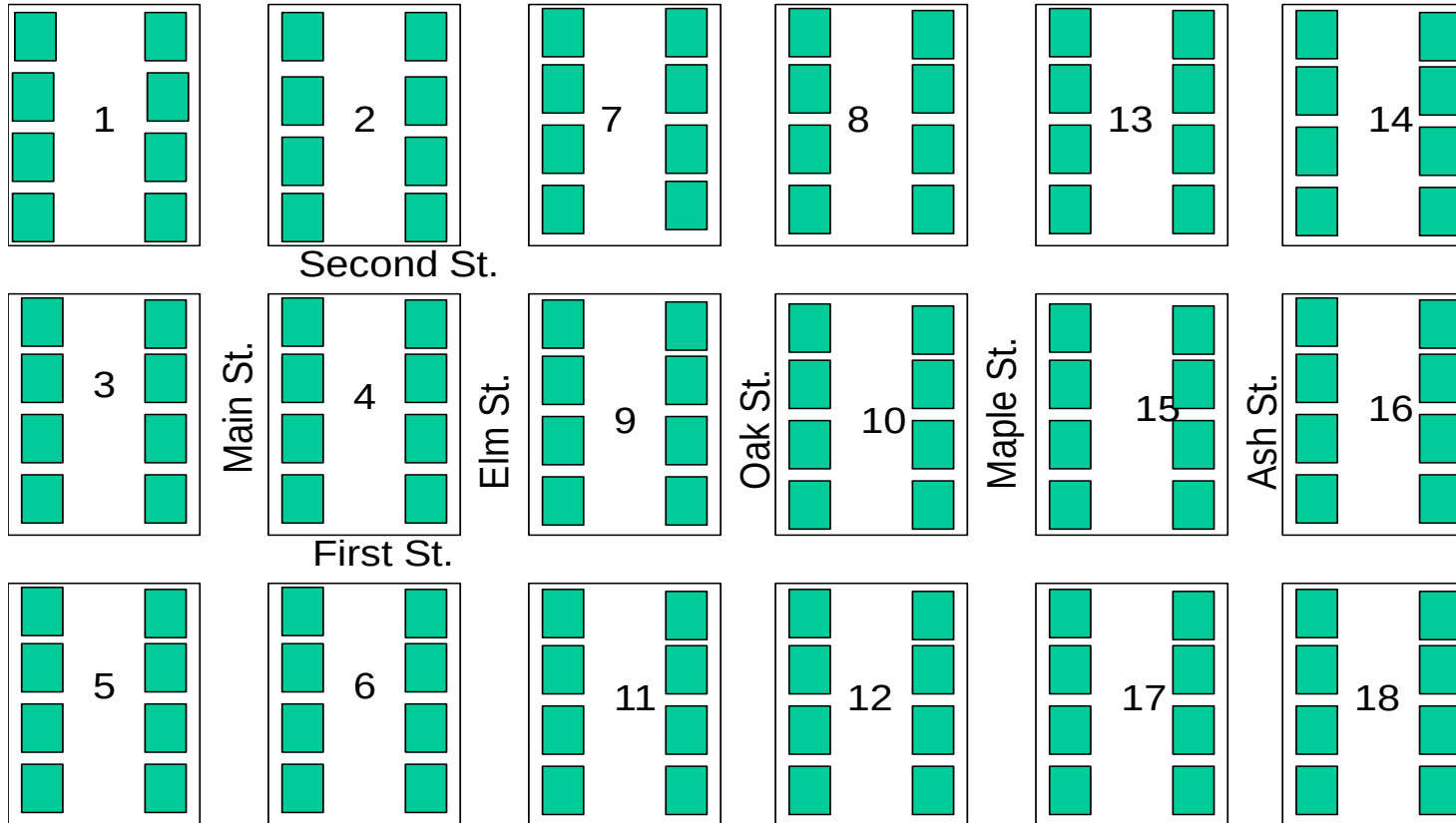
- SRS of clusters and SRS of elements within
- Define the average within cluster variance as $S_b^2 = \frac{1}{A} \sum_{\alpha=1}^A S_{\alpha}^2$
- Sampling variance changes:

$$Var(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{S_a^2}{a} + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{ab}$$

Clustered population

$$\bar{Y}_\alpha = \sum_{\beta=1}^B Y_{\alpha\beta} / B$$

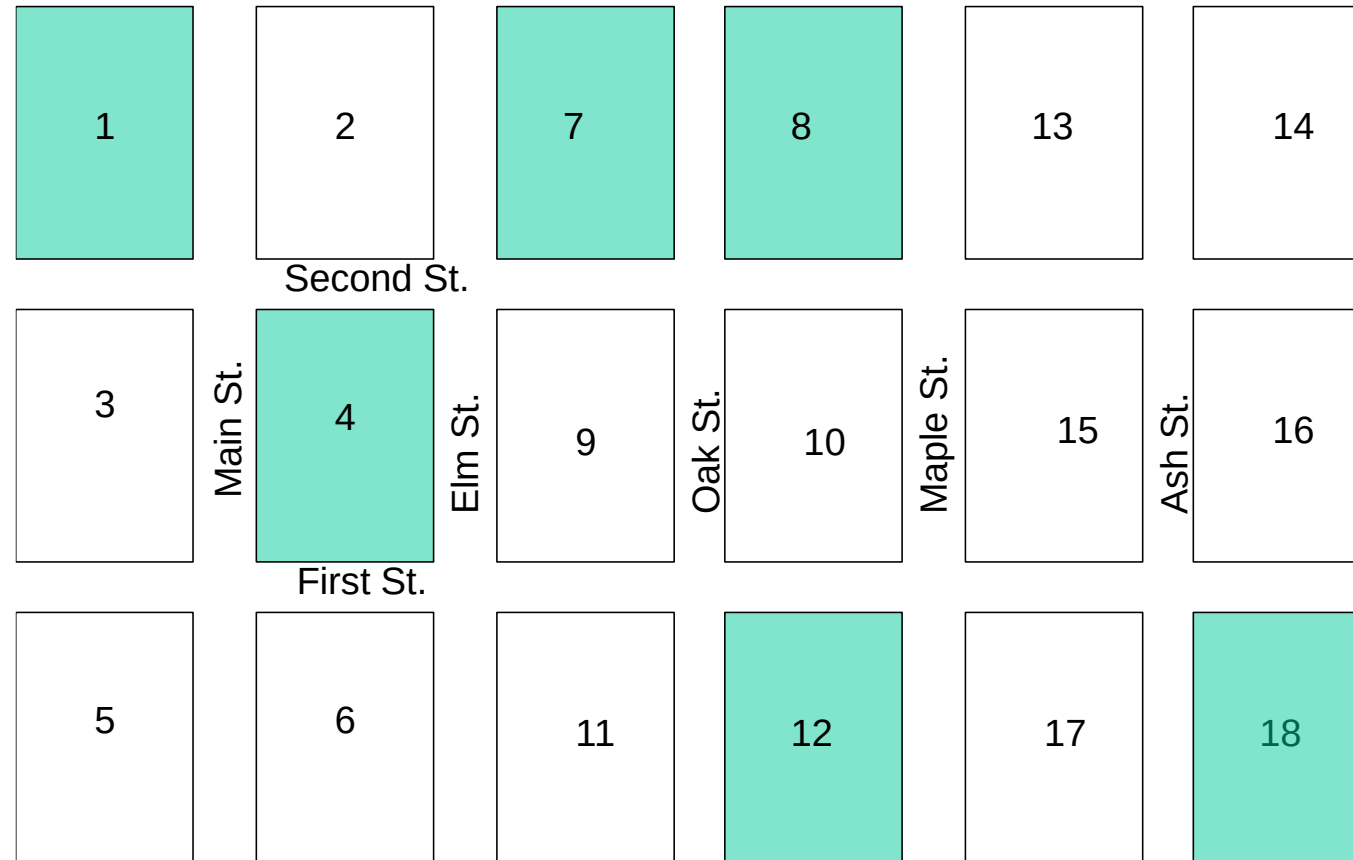
$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$S_\alpha^2 = \sum_{\beta=1}^B (Y_{\alpha\beta} - \bar{Y}_\alpha)^2 / (B-1)$$

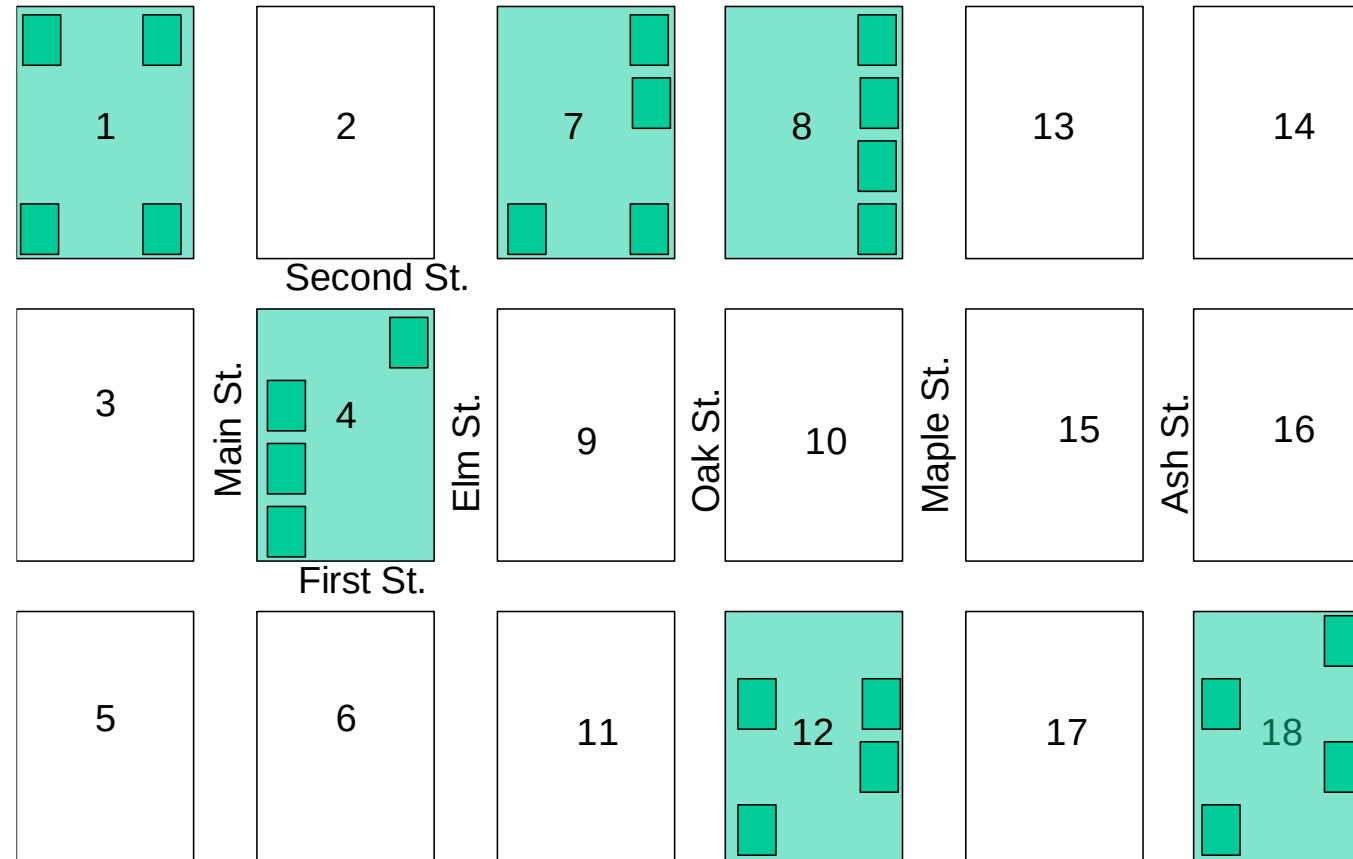
$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Two-stage cluster sample: 1st stage



Two-stage cluster sample: 1st and 2nd stage

$$\bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta} / ab$$



$$Var(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{S_a^2}{a} + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{ab}$$

Estimation of sampling variance

- Estimation of $Var(\bar{y})$ is complicated by the estimation of S_a^2

$$E(s_b^2) = S_b^2$$

$$E(s_a^2) = S_a^2 + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{b}$$

$$\text{var}(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{s_a^2}{a} + \frac{a}{A} \left(1 - \frac{b}{B}\right) \frac{s_b^2}{ab}$$

Approximation

- If a/A is negligible, then $\text{var}(\bar{y})$ reduces to

$$\text{var}(\bar{y}) = \frac{1}{a} s_a^2$$

- This approximation avoids the need to compute s_b^2
- There is an alternative conceptual explanation for using a reduced or simplified variance estimate ...
- Ultimate clusters
- Leads to yet a second approximation: $\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$

Practical sampling variance estimation

- If clusters are selected with replacement and elements without, use the unbiased estimate:

$$\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

- If clusters are selected without replacement and element without, use a slightly biased estimate:

$$\text{var}(\bar{y}) \approx \frac{(1-f)}{a} s_a^2$$

instead of the more complicated

$$\text{var}(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{s_a^2}{a} + \left(\frac{a}{A}\right) \left(1 - \frac{b}{B}\right) \frac{s_b^2}{ab}$$

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B. Sample size and *roh*

- **Effective sample size**: size of SRS that would give the same precision as the cluster sample
- $n_{eff} = n/deff$
 - $deff = 2.0$ and $n = 1500$ then $n_{eff} = 1500/2 = 750$
- Varying subsample size for fixed n changes the number of clusters, $deff$, and the effective sample size
- For two-stage cluster sample: $deff = 1 + (b - 1)roh$

Effective sample size

- Cluster sample with $n = 2000$, and $roh = 0.03$:
- For $b = 10$, $deff = 1 + (10 - 1)0.03 = 1.27$
 - $n_{eff} = 2000/1.27 = 1574.80$
- For $b = 20$, $deff = 1 + (20 - 1)0.03 = 1.57$
 - $n_{eff} = 2000/1.57 = 1273.89$
- For $b = 50$, $deff = 1 + (50 - 1)0.03 = 2.47$
 - $n_{eff} = 2000/2.47 = 809.72$

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C. Homogeneity and *roh*

- Sample 10 school classrooms from 1,000
 - Each classroom has exactly 24 children
- Alternative 1: sample characteristic is dichotomy

$$\frac{9}{24}, \frac{11}{24}, \frac{13}{24}, \frac{15}{24}, \frac{16}{24}, \frac{17}{24}, \frac{18}{24}, \frac{20}{24}, \frac{20}{24}, \frac{21}{24}$$

- Here the intra-class correlation $roh = 0.088$
- Moderate amount of homogeneity within clusters
- $deff = 1 + (24 - 1)0.088 = 3.024$
- Actual sample size $n = 240$
- Effective sample size $n_{eff} = 240/3.024 = 79.37$

Perfect homogeneity

- Homogeneity within, heterogeneity among:

$$\frac{0}{24}, \frac{0}{24}, \frac{0}{24}, \frac{16}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}$$

$$deff = 23.90$$

$$roh = \frac{23.90 - 1}{24 - 1} = 0.996$$

$$n_{eff} = 240 / 23.9 = 10.04$$

Perfect heterogeneity

- Heterogeneity within, homogeneity among:

$$\frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}$$

$$deff = 0$$

$$roh = \frac{0-1}{24-1} = -0.04348$$

$$n_{eff} = 240 / 0 = \text{undefined (total population)}$$

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D. Portability of *roh*

- Estimation (1)

$$\text{var}_{(1)}(p) = \frac{(1-f)}{a} s_a^2$$

$$\text{var}_{(1),SRS}(\bar{y}) = \frac{p(1-p)}{n_{(1)}}$$

$$\text{deff}_{(1)} = \frac{\text{var}_{(1)}(p)}{\text{var}_{(1),SRS}(p)}$$

$$\text{roh} = \frac{\text{deff}_{(1)} - 1}{b_{(1)} - 1}$$

- Design (2)

$$\text{var}_{(2)}(p) = \text{deff}_{(2)} \times \text{var}_{(2),SRS}(p)$$

$$\text{var}_{(2),SRS}(\bar{y}) = \frac{p(1-p)}{n_{(2)}}$$

$$\text{deff}_{(2)} = 1 + (b_{(2)} - 1)\text{roh}$$

roh

Illustration

- Crime victimization survey in a large public housing project
- Sample apartments
- Responsible adult interviewed about victimizations occurring to any member of the household
- $A = 400$ floors with exactly $B = 15$ apartments
- $a = 10$ floors selected with replacement
- $b = 5$ apartments selected without replacement

Sample results

$$f = \frac{a}{A} \times \frac{b}{B} = \frac{10}{400} \times \frac{5}{15} = \frac{50}{6000} = \frac{1}{120}$$

Sample floor	HH's experienced any crime (y_α)
1	4
2	4
3	5
4	5
5	3
6	1
7	0
8	1
9	2
10	1

Estimation

$$p = \bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta} / n = 26 / 50 = 0.52$$

$$\text{var}(\bar{y}) = \frac{1-f}{a} s_a^2$$

$$s_a^2 = \frac{1}{b^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right] = \frac{1}{(5)^2(9)} \left[98 - \frac{(26)^2}{10} \right] = 0.13511$$

$$\text{var}(p) = \frac{1 - \frac{1}{120}}{10} (0.13511) = 0.013399$$

$$se(p) = 0.11575$$

Confidence interval, *deff* and *roh*

$$t_{(0.975, a-1)} = t_{(0.975, 9)} = 2.262$$

$$p \pm t_{(0.975, 9)} se(p) = (0.25815, 0.78185)$$

$$\begin{aligned} \text{var}_{srs}(p) &= (1-f) \frac{p(1-p)}{n-1} = \\ &= \left(1 - \frac{1}{120}\right) \frac{(0.52)(0.48)}{49} = 0.005051 \end{aligned}$$

$$deff = \frac{\text{var}(p)}{\text{var}_{srs}(p)} = \frac{0.013399}{0.0050514} = 2.6524$$

$$roh = \frac{deff - 1}{b - 1} = \frac{2.6524 - 1}{5 - 1} = 0.41311$$

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E. Subsample size

- Precision of estimates depends on a and b
- Optimum choice for a and b can be obtained by minimizing the sampling variance for fixed cost (or vice versa)
- Cost model for two stage sampling:
 - $C - C_0 = a \times c_a + a \times (b \times c_b)$
 - C_a is the cost per cluster and C_b is the cost per observation within a cluster
- C_b is dominated by interviewing costs
- C_a is dominated by travel and preparation costs

Optimization

- Optimum subsample size for fixed cost $C - C_0$ can be found using a Lagrange multiplier or the Cauchy-Schwartz inequality
- Find a (or b) to minimize the sampling variance

$$Var(\bar{y}) = \frac{S_a^2}{a} + \frac{S_b^2}{ab}$$

- The optimum b is

$$b_{opt} = \sqrt{\frac{c_a}{c_b} \cdot \frac{1 - roh}{roh}}$$

Illustration

- Large national personal visit interview survey
- Interviews conducted within respondent homes by locally hired interviewers
- *roh* values near 0.05
- $n = 1,000$ is fixed
- 1 hour interview requires 2 hours interviewer time
 - Interviewer travels to the cluster, locating households, making callbacks, etc.
 - At \$12.50 per interviewer hour, $c_b = 2 \times \$12.50 = \25

Cost per cluster

- Household clusters (segments) an average 1/2 hour drive from interviewer homes
- Require an average of 3 visits each
- $3 \times \$12.50 = \37.50 required to pay interviewers for travel time
- Travel averages 30 m.p.h.
- At \$0.31 per mile, 6 trips of one half hour (15 miles) requires travel reimbursement of $(6 \times 15)(0.31) = \$27.90$
- $c_a = 37.50 + 27.90 = \$65.40$

Optimum b

- The optimum subsample size will be

$$b_{opt} = \sqrt{\frac{65.40}{25} \cdot \frac{1-0.05}{0.05}} = 7.05$$